

FIFTH EDITION

ESSENTIALS of STRENGTH TRAINING and CONDITIONING

G. Gregory Haff
N. Travis Triplett
editors



Includes exercise
technique videos
with HKPropel
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ESSENTIALS OF STRENGTH TRAINING AND CONDITIONING

FIFTH EDITION



G. Gregory Haff, PhD, CSCS,*D, FNSCA
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EDITORS



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PREFACE

In 1994, the first edition of *Essentials of Strength Training and Conditioning* was published. After a second edition (in 2000) and sales of over 100,000 books, an expanded and updated third edition was published in 2008. This newest edition continues the tradition as the most comprehensive reference available for strength and conditioning professionals. In this text, 30 expert contributors further explore the scientific principles, concepts, and theories of strength training and conditioning and their applications to athletic performance.

The first edition grew out of an awareness that there was not a book about strength training and conditioning that captured the views of leading professionals in anatomy, biochemistry, biomechanics, endocrinology, nutrition, exercise physiology, psychology, and the other sciences and that related the principles from these disciplines to the design of safe and effective training programs. Also, the lack of relevant and well-conducted research studies had hindered earlier efforts to create an all-inclusive resource. Once it was finally developed, *Essentials of Strength Training and Conditioning* quickly became the definitive textbook on the subject.

The second edition, released six years later, was more than a simple freshening of the content; it was an overhaul of the scope and application of the first edition. Throughout the text and in the additional 100-plus pages, the chapter contributors used updated, relevant, and conclusive research and concepts to turn scientific information into information on performance. Many learning tools were added, such as chapter objectives, key points, application

boxes, and sample resistance training programs for three different sports. These enhancements, plus the addition of a full-color interior and hundreds of color photographs, made the second edition truly exceptional.

The third edition, released eight years after the second edition, offered restructured chapters and expansions of other chapters complete with new photographs and updated terminology. In addition, the artwork was modernized and instructor and student resources were created to help keep this text the primary resource for the study and instruction of strength and conditioning.

The fourth edition—published seven and a half years after the third—expanded on the previous editions by incorporating the most current research and best practices available at the time. Notable updates were made in content related to high-intensity interval training, overtraining, agility and change of direction, nutrition for health and performance, and periodization. A new chapter was added featuring instructions and photographs for exercises using alternative modalities and nontraditional implements. In addition, an online exercise technique video library was introduced to support instructors, students, and professionals who may not have access to a training facility. Finally, ten new tests—including several assessing maximal strength, power, and aerobic capacity—were added, along with updated flexibility movements, resistance training exercises, plyometric drills, and speed and agility exercises, enabling readers to design programs aligned with contemporary guidelines.

Updates to the Fifth Edition

This fifth edition remains the single best resource for the key theories, concepts, and scientific principles of strength training and conditioning. Its scope, precision, and dependability make it a definitive reference for strength and conditioning professionals to consult for information on the structure and function of body systems, training adaptations, testing and evaluation, exercise technique, program design, and the organization and administration of facilities. The new edition made these updates and enhancements:

- Expanded exercise and sport applications related to the anatomical planes of movement
- Introduced guidelines for promoting mental health and understanding the psychological impact of sport injury
- Updated information on performance-enhancing substances and methods
- Incorporated new tests and normative and descriptive data tables for assessing athletic populations
- Included additional exercises and drills for dynamic stretching, resistance, plyometric, speed, agility, and aerobic training
- Outlined common errors for flexibility exercises, plyometric drills, and aerobic endurance machines
- Provided verbal cues and common errors for resistance training exercises
- Doubled the number of alternative mode and nontraditional implement exercises, plus speed and agility drills
- Supplied new variations, regressions, and progressions for speed and agility drills
- Updated athlete scenarios for resistance training programming
- Introduced a sample 12-month periodized training plan
- Presented guidelines for aerobic endurance and metabolic training using five- and three-zone intensity scales

- Added information on rhabdomyolysis, sickle cell trait, heat illness, and cold exposure
- Developed an all-new chapter on overreaching, overtraining, and recovery, including strategies to reduce risk
- Updated facility layout, design, and maintenance guidelines to reflect modern strength and conditioning environments
- Described common litigation issues related to facilities and professional practice, with strategies to minimize liability

Instructor Resources in HKPropel

The new edition includes updated and expanded instructor resources, which are available within the instructor pack in HKPropel:

- *Instructor Video.* The instructor video collection contains 100 supplemental clips demonstrating correct and incorrect tests, and resistance, plyometric, and alternative mode exercises. These videos are invaluable for instructors who need to teach proper exercise technique when there is a lack of instructional experience, facility equipment, space, or access, or when there's a combination of these issues.
- *Instructor Guide.* The instructor guide contains a course description, a sample semester schedule, chapter objectives, chapter outlines, key terms with definitions, and a list of websites and additional resources.
- *Presentation Package and Image Bank.* This comprehensive resource, delivered in Microsoft PowerPoint, offers instructors a presentation package containing over 600 slides to help augment lectures and class discussions. In addition to outlines and key points, the resource contains most of the figures, content photos, and tables from the text, sorted by chapter. These can be used in developing a customized presentation based on specific course requirements.
- *Test Package.* The test package includes a bank of 780 questions in true-false and multiple-choice formats. The files may be downloaded for integration with a learning management system or printed for use as paper-based tests. Instructors may also create their own customized quizzes or tests from the test bank questions to assign to students directly through HKPropel.
- *Chapter Quizzes.* Contains ready-made quizzes (10 questions each) to assess student comprehension of the most important

concepts in each chapter. Each quiz may be downloaded or assigned to students directly through *HKPropel*.

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The online content includes a collection of video clips of 25 resistance training exercises for use in understanding and performing correct exercise technique. Plus, lab activities are provided to give students hands-on practice with testing and evaluation. The 20 forms make completing and submitting lab assignments easy.

References

The complete list of references for *Essentials of Strength Training and Conditioning, Fifth Edition*, is available online only. It is available to instructors, students, and professionals in HKPropel, and the references are also available with no access code needed at <https://ancillaries.humankinetics.com/EssentialsOfStrengthTrainingAndConditioning5thEdition> and at the QR code here.



Certification Exams

Essentials of Strength Training and Conditioning is the primary resource for individuals preparing for the National Strength and Conditioning Association's Certified Strength and Conditioning Specialist (CSCS) certification exam.

As a worldwide authority on strength and conditioning, the National Strength and Conditioning Association (NSCA) supports and disseminates research-based knowledge and its practical application to improve athletic performance and fitness. With over 60,000 members and certified professionals, the NSCA has established itself as an international clearinghouse for strength and conditioning research, theories, and practices.

The NSCA offers five distinct certifications that serve the full spectrum of fitness, performance, and tactical professions. The Certified Strength and Conditioning Specialist (CSCS) supports coaches who train athletes and teams by emphasizing scientific principles, performance testing, and sport-specific program design. The NSCA Certified Personal Trainer (NSCA-CPT) equips trainers who work with general-population clients to conduct assessments, plan individualized programs, and ensure safe, effective exercise technique. The Certified Special Population Specialist (CSPS) focuses on professionals who train clients with chronic conditions or unique health needs, highlighting tailored modifications and safety-focused programming. The Tactical Strength and Conditioning Facilitator (TSAC-F) prepares those who support military, law enforcement, and first-responder personnel by emphasizing physical readiness, injury-risk reduction, and resilience. Finally, the Certified Performance and Sport Scientist (CPSS) is designed for practitioners who apply research, data, and technology to monitor, evaluate, and enhance athletic performance. All five NSCA credentials follow rigorous professional standards and require ongoing continuing education to remain active.

CSCS and NSCA-CPT were the first certifications of their kind to be nationally accredited by the National Commission for Certifying Agencies, a nongovernmental, nonprofit agency in Washington, DC, that sets national standards for certifying agencies.

Whether used for learning the essentials of strength training and conditioning, for preparing for a certification exam, or as a reference by professionals, *Essentials of Strength Training and Conditioning, Fifth Edition*, will help practitioners and the scientific community better understand how to develop and administer safe and effective strength training and conditioning programs.

ACCESSING THE LAB ACTIVITIES

The lab activities are accessed through *HKPropel*.

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Lab 1: Anaerobic Capacity Testing

300-Yard (274 m) Shuttle Run

Lab 2: Aerobic Capacity Testing

1.5-Mile (2.4 km) Run

12-Minute Run

Lab 3: Anthropometry and Body Composition

Skinfold Measurements

Lab 4: Exercise Testing for Athletes

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Lab 9: Speed and Agility Technique and Testing

T-Test

Hexagon Test

5-10-5 Test

40-Yard (37 m) Sprint

Lab 10: Muscular Endurance Testing

Push-Up Test

YMCA Bench Press Test

Partial Curl-Up Test

Lab 11: Facility Layout Design

Facility Floor Plan

ACKNOWLEDGMENTS

The development of the fifth edition of the NSCA's *Essentials of Strength Training and Conditioning* was a massive undertaking that would not have been possible without the contributions of a vast number of people. The historic development of this iconic text has served as our guiding principle, and the hard work of the numerous authors who contributed to the four previous editions has established a strong foundation for this text. Therefore, we thank the previous editors, Thomas Baechle and Roger Earle, for their foresight over thirty years ago that has led us to where we are today and for their passionate work on the previous editions. This edition would not have been possible without the continued contribution of Roger Earle, who has gone beyond his role as a Human Kinetics representative. He is a true friend who has helped with many aspects of this book and our writing careers.

We would also like to thank Keith Cinea for his help throughout the process. Because it is a key resource for current and future strength and conditioning professionals, it was essential for us to ensure that this text holds true to the NSCA mission of translating science into practice, and Keith is an ambassador of this philosophy. Thanks also to the multitude of individuals at Human Kinetics who were essential to completing every phase of the publication of this book, from copyediting to graphic design.

G. Gregory Haff, PhD, CSCS,*D, FNSCA

To my coeditor and life-long friend, Travis Triplett: It is hard to believe all that we have accomplished across our careers and with this project. While editing a book of this magnitude is challenging and at times very frustrating, your ability to remain positive and focused on the end goal is inspiring. The balancing act of editing this book can only be accomplished with two people that complement each other's strengths and believe in the NSCA. I can think of no one other than you who I would want to attempt a project like this with. Your kind heart and easygoing style is a perfect complement to my "bull in a China shop" methodology for processes like this. Thanks for always being one of my very best friends, and the next margarita is on me!

I must thank my family. My wife Erin who has been by my side for well over two decades pushing me to be the best I can be and providing me the love and support that I need to continue to chase my dreams and undertake projects like this. Without her support I would merely be stuck under the heavy lifting bar of life. It is a blessing to have someone strong enough to spot you when times are tough, and for that I love you more than you know. For my father, Guy Haff; the US Marine Corp values you instilled within me from a young age have served as the foundation for all things that I do and after all these years I truly get why you were disappointed that I did follow in your footsteps into the Marine Corp. It is hard to believe the training that you had me do at the West Morris YMCA at 11 years of age would set the foundation for my life's work. Finally, I must dedicate my efforts to my mother, Sandra Haff who pushed me to read as many books as I could to overcome my reading deficiencies, to always be physically ready for whatever is around the corner, and to fight for what is right no matter the cost. No matter where you are now, I hope you are still proud of the man I am and the man I strive to be each and every day. I miss you much, Mom,

and I wish you were here to see all the great things that have happened.

N. Travis Triplett, PhD, CSCS,*D, FNSCA

I never dreamed that taking my first weight training class while at the university would have culminated in such a rewarding career in the field of strength and conditioning. It is difficult to thank every person who had a role in getting me to this point in my life and my career, which enabled me to enthusiastically embark on this project. I was fortunate to receive a strong foundation from my parents—I wish you could both be here to see that the example you set was followed. I also want to thank my brother and my circle of friends, both personally and professionally, who have always been supportive and have been there to brighten my day. While the list of individuals who have influenced my career is long, my two greatest guides have been Mike Stone and Bill Kraemer. I value your mentorship and friendship greatly. Numerous colleagues and former students around the world have contributed to my knowledge and success along the way, and I appreciate each and every one of you even if we don't see each other very often.

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